

Project Report

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ADTA 5560: Recurrent Neural Network with Sequence Data

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Using Deep Learning for Weekly Crime Forecasting in Chicago: Evidence from 20 Years of Data

Abstract

Predicting urban crime is crucial to ensure the safety of the community by allocating law enforcement resources more efficiently. This study utilizes 20 years of crime data from the city of Chicago to run and evaluate various Recurrent Neural Network (RNN) architectures such as Vanilla RNN, Bidirectional RNN, LSTM, and GRU against the traditional ARIMA statistical model. This was achieved by aggregating 8.5 million records into weekly counts that capture long-term downward trends and strong seasonal fluctuations. Our findings show us that deep learning models, particularly the GRU model outperform traditional methods in capturing non-linear temporal dependencies providing a better insight for short term operational planning in public safety.

Literature Review and Problem Statement

Crime forecasting has been increasing attention among different disciplines. Spatiotemporal prediction and analysis of crime are considered as one of the important monitoring tools ([Kounadi et al., 2020](#)). Researchers who conducted a study in this area used forecast crime across time and space, and their analysis presents that crime does not happen randomly, but has some spatial and temporal patterns that can be modeled and predicted. Crimes pose a significant threat to the safety of residents and have a great impact on social and economic stability. Therefore, forecasting crime plays a key role in improving public safety and policymaking. Their accurate predictions help to allocate resources to the places with real needs ([Jenga et al., 2023](#); [Agarwal et al.](#)). These predictions can save lives, prevent property damage, and improve the effectiveness of resources.

Access to historical data provides an opportunity to forecast future crimes ([Jenga et al., 2023](#)), and many of the prediction methods used retrospective data to forecast future events. These models are the foundation of modern crime analysis, and past crime occurrences are used to estimate future risks. However, crime forecasting is inherently spatiotemporal ([Kounadi et al., 2020](#)), and environmental and spatial characteristics can influence the risk of higher crimes. These

characteristics can affect the built environment of a city, including road networks, visibility, lighting, and land use ([Ahmadi & Taleai, 2025](#)). These findings present the need to combine spatial variables into forecasting models to identify hotspot areas with more prevalent crimes.

Analyzing crime was done first using traditional statistical models and then advanced machine learning models. Early methods focused on time series statistical methods and used spatial maps to find patterns from historical crime data to predict future trends. By the advancement of data mining and machine learning, the methodologies used for crime prediction have also been improved. One of the common methods used in forecasting crime is the Autoregressive Integrated Moving Average (ARIMA) model. Among these studies, [Islam and Raza \(2020\)](#) deployed ARIMA to predict crime levels for the next two years. Their findings showed that ARIMA has better performance in comparison to exponential smoothing with about 80 percent accuracy. Similarly, [Ghani et al. \(2023\)](#) utilized ARIMA to create choropleth maps for prediction of crimes. They integrated ARIMA modeling with geographic tools such as QGIS to generate spatial maps and to display hotspot areas. This method was known as a primary form of predicting crimes which uses retrospective data to find regions with high crime events ([Chainey et al., 2008](#)).

Later, with increased availability of data and computational power, machine learning models have been introduced to enhance predictive accuracy. For example, [Tamir et al. \(2021\)](#) deployed various machine learning models to predict the crimes in Chicago. They used Random Forest, K-Nearest Neighbors (KNN), AdaBoost, and neural networks. Among these models, neural networks could reach the highest performance with 90.77 percent accuracy. Their results showed that RNN was better at prediction than traditional methods. Also, [İlgün and Dener \(2025\)](#) used data related to crimes in Chicago, San Francisco, and Philadelphia to design different models, including XGBoost, CatBoost, Random Forest, Decision Trees, Multilayer Perceptron (MLP), KNN, Gaussian Naive Bayes, and Logistic Regression.

Deep learning models also demonstrated better performance when they were compared with traditional methods such as ARIMA. These deep learning models are capable of capturing long-term and nonlinear relationships. [Muthamizharasan and Ponnusamy \(2024\)](#) tried to compare ARIMA and LSTM models for crime event forecasting. Their results showed that the ARIMA method is just suitable for conventional time-series forecasting, and it cannot find complex and

nonlinear relationships in crime data. However, LSTM outperforms traditional methods in finding sequential and long-term dependencies.

Similarly, [Wang et al. \(2017\)](#) tried to address the challenges of handling crime prediction by using a spatiotemporal deep learning framework (ResNet) to predict crime distribution in Los Angeles. Their findings demonstrated that this type of model had highly accurate predictive power. [Stec and Klabjan \(2018\)](#) also reported the efficiency of deep neural networks for prediction of crime on the next time step using Chicago and Portland datasets by combining external predictors such as weather, census data, and public transportation. Deep learning also was utilized in multimodal crime forecasting. [Kang and Kang \(2017\)](#) integrated crime counts with demographic, meteorological, and environmental data and they used shared feature layers.

Besides that, [Butt et al. \(2024\)](#) used transfer learning in deep crime prediction. They used LSTM and CNN-LSTM in their study, and they proposed a transfer learning BiLSTM model. Their model transferred crime information from one area to another close region and achieved good performance in Chicago, New York, and Lahore datasets. In addition to these studies, hybrid methods that integrate statistical models and neural networks could receive a lot of attention among researchers. For example, [Soares et al. \(2020\)](#) developed a hybrid SARIMA and ANN model to predict crime rates, and their results showed that a combination of linear and nonlinear models can have better efficiency. Similarly, Iacobescu and [Iacobescu and Susnea \(2025\)](#) tried to compare a hybrid ARIMA-ANN model with traditional classification techniques. Their findings revealed that this model outperforms traditional approaches in the identification of seasonal and temporal patterns.

Even though previous studies proved the capability of deep learning models such as LSTM, some focused on forecasting the next day or next month, and they lack the ability to perform predictions in several future time steps. Besides that, they evaluate the performance of the models over limited time windows. So, using historical data over 20 years is a great benefit for the development of a model that can cover structural changes over a long period of time. So, the main goal of this study is to use 20 years of crime data in Chicago, which provides an appropriate volume of data at the week level, to propose and develop an RNN model that can predict these data for future weeks ahead. The result of this study is a paramount step in the prediction of criminal events and

accommodates the public safety systems to have better control over the regions (Shah et al., 2021). So, this study will be guided by the following research questions and dataset.

Data Introduction

For this project, we used City of Chicago Crime Data (Chicago Data Portal, 2026), which is the official archive maintained by the Chicago Police Department. It covers a period of 20 years of data on reported crime in Chicago. Each entry gives us a detailed look at an incident, including the type of crime, when it happened, and the general city block where it took place. This study will focus on weekly crime count using long-term historical data.

Target Variable

The target variable of this study can be defined as the total number of crimes per week. Since we want to forecast broader patterns rather than pinpointing single events, we’ve "zoomed out" by aggregating the raw Chicago dataset, which reports crime in a daily manner, into weekly crime counts (total number of reported incidents across the city in any given seven-day window). This turned a massive list of events into a structured timeline of about 1,200 weeks, which will be perfect to cancel out any noise we may have encountered for the daily count as well as provide the RNN enough records to learn the patterns in a meaningful way, as opposed to if it was a monthly count, where the number of records would be under 300, which won’t be sufficient for the RNN to learn anything of substance.

Data collection and Characteristics

The crime data of Chicago was obtained from the Chicago Data Portal and the City of Chicago Crime Data. It contains 8497120 records with 22 fields. Detailed information about all the features of this dataset is presented in Table 1.

Table 1

Data Dictionary of the Chicago City Crime Dataset

Column	Data Type	Range / Unique Info	% Missing	Description
ID	int64	634 – 14,038,975	0.00%	Unique numeric identifier for each crime record.

Case Number	object	1,048,014 unique	0.00%	Official police case number assigned to the incident.
Date	object	407,938 unique timestamps	0.00%	Date and time when the crime occurred.
Block	object	54,941 unique	0.00%	Approximate street block location of the crime (exact address masked).
IUCR	object	382 unique	0.00%	Illinois Uniform Crime Reporting code (detailed offense classification).
Primary Type	object	34 unique	0.00%	Main crime category (e.g., THEFT, BURGLARY).
Description	object	477 unique	0.00%	Detailed description of the offense.
Location Description	object	212 unique	0.15%	Type of place where the crime occurred (e.g., STREET, RESIDENCE).
Arrest	bool	True / False	0.00%	Indicates whether an arrest was made.
Domestic	bool	True / False	0.00%	Indicates whether the crime was domestic-related.
Beat	int64	111 – 2535	0.00%	Police patrol beat (smallest police geographic unit).
District	int64	1–31	0.00%	Police district number.
Ward	float64	1 – 50	56.57%	Chicago city council ward number.
Community Area	float64	0 – 77	56.34%	Chicago community area (77 official zones).
FBI Code	object	26 unique	0.00%	FBI crime classification code.
X Coordinate	float64	0 – 1,205,119	2.37%	Projected east–west coordinate (Illinois State Plane system).
Y Coordinate	float64	0 – 1,951,622	2.37%	Projected north–south coordinate.

Year	int64	2001–2025	0.00%	Year extracted from the Date column.
Updated On	object	3,300 unique	0.00%	Date and time when the record was last updated.
Latitude	float64	36.62 – 42.02	2.37%	Geographic latitude (WGS84).
Longitude	float64	-91.69 – -87.52	2.37%	Geographic longitude (WGS84).
Location	object	436,891 unique	2.37%	Combined (Latitude, Longitude) string representation.

Note. This table presents the names of fields and their corresponding data types. It shows the range of values in each column and the percentage of missing values we need to preprocess in each field.

Why was this project selected?

We selected the Chicago crime dataset because it provides us with 20 years of time stamped incident data which is perfectly suited for a time series forecasting using Recurrent Neural Network (RNN). Simple statistical models would not be able to gauge things like seasonality, trends, and fluctuations properly. The size of the dataset and its long time period make it perfect for a forecasting analysis with RNN. The data will be aggregated to “weekly crime count,” which will give us a sufficient sequence length for neural network training and reduce daily noise while improving temporal pattern learning. The suitability for similar crime data for pattern detection and predictive modeling has been supported in prior research ([Wang et al., 2013](#))

Why is it important?

Crime forecasting is a crucial problem that directly links to public safety, logistics, and strategic planning in urban environments. Being able to accurately predict future crime trends based on factors like time periods and locations will in turn help promote better policing and allocation of law enforcement resources. Weekly crime forecasts are especially beneficial for short-term operational planning and resource allocation. Understanding long-term crime patterns will also help policymakers design stronger crime prevention strategies based on real data, specifically targeting core areas of conflict. This analysis will demonstrate how deep learning is very useful to extract temporal dependencies from a large-scale time series dataset. The importance of predictive crime analysis and, by extension of that, better policymaking and public safety has also been well documented in prior research ([Kang & Kang, 2017](#)).

Who is our core audience/who does it matter to?

The project is important to multiple stakeholders. Firstly, law enforcement agencies can use the forecast to optimize their patrol planning as well as reduce response time for any freshly occurring incidents. Secondly, policymakers and city planners can use this information to design safer communities and allocate safety budgets more efficiently. Thirdly, data scientists and analysts, especially those who study crime patterns, benefit from the added information made by RNN-based forecasts to boost their research. Last but not least, residents also benefit from the better implemented safety protocols based on the report as well as general awareness. It is important to note that weekly forecasts provide more operationally relevant insights, as policymaking, strategies, and deployment are often planned on a weekly basis. There has been a strong relevance of crime forecasting to law enforcement agencies and policymakers in public safety planning that is widely recognized in crime prediction research ([Shah et al., 2021](#)).

Methodology

First, we need to generate a dataset from the available original dataset that is clean and suitable for model development. So, as shown above, the original crime dataset includes many fields, and each row is representative of a single crime that happened on a particular day in a particular location or community. Besides that, the type of crime that occurred is also specified. For this project, we decided to consider the simple scenario and just focused on the number of crimes that happened in each week. Therefore, we ignored using other factors such as geographic location or crime group type or other attributes.

Accordingly, our goal is to build a time series model using only temporal information. So, we created a reduced dataset from the initial one that contains two fields, including the week that shows the timestep and total number of crimes that happened in each week. So, since we don't use other features, we didn't clean all of those fields as well. This dataset was obtained from the API provided by the website, and we downloaded the most recent accessible dataset for Chicago crime in 2026, which was retrieved in batches of 100,000 records. This dataset includes 8,537,223 rows and 22 fields. Then based on the aforementioned preprocessing, this dataset was then transformed into a weekly aggregated format appropriate for modeling.

Data Preprocessing

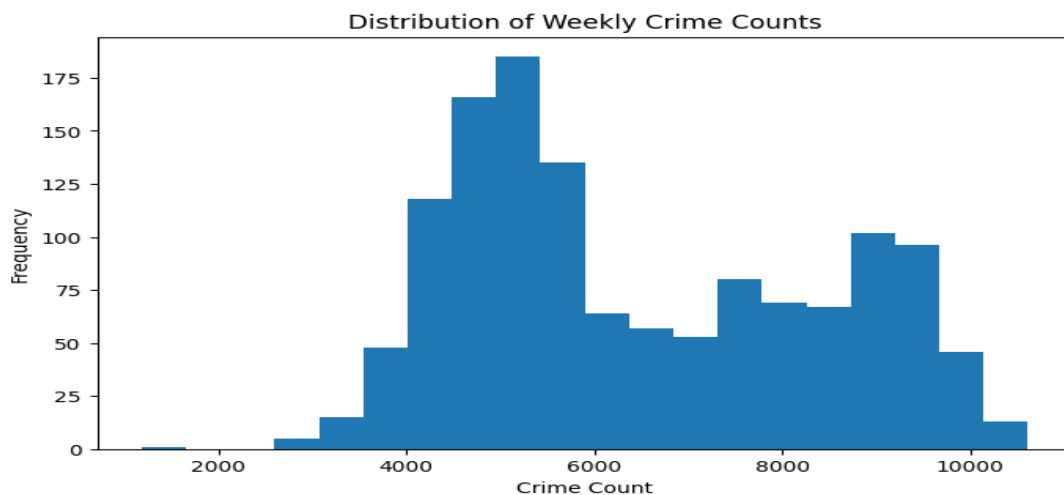
In this stage we tried to prepare the raw data for time series modeling and transformed it into a clean weekly format. For this purpose, we first convert the field that contains date information into

a clean weekly format. So, any dates that are invalid or do not follow the correct format were identified and then they were converted into null values, and then they were removed from the dataset. In the next step, each of those dates was mapped to its related week. This helps to aggregate group daily records into weekly terms. Then, these weekly periods were converted into real time steps that showed the start of each week, and these values were stored in a new field called "week". Subsequently, the number of crimes for each week was computed and stored in a field called "crime count" The result is a preprocessed and clean dataset for time series modeling.

Exploratory Data Analysis

For understanding the new dataset and its statistical distribution, besides finding other patterns and outliers, we conducted exploratory data analysis. So, first we created a histogram for the crime count. This histogram illustrates that the distribution of weekly crime counts over the observed period is not normally distributed, and it has two modes, and it appears to have two common ranges when crime counts frequently occurred. So the values first were concentrated around 4500 to 6000 crimes per week, and the second observable peak shows that they were around 8000 to 9500 crimes per week. This pattern can have some specific meaning, and it shows the differences between the earlier and recent years, or some changes probably happened relevant to social or political regulations (Figure 1).

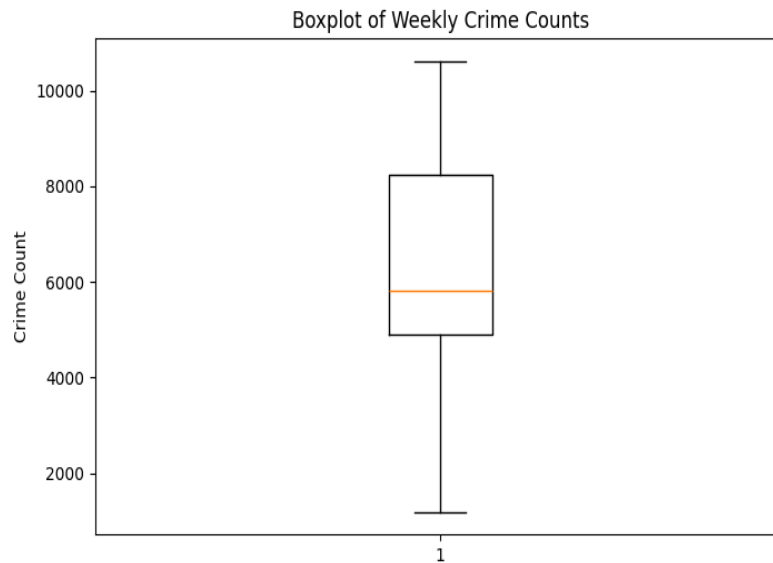
Figure 1



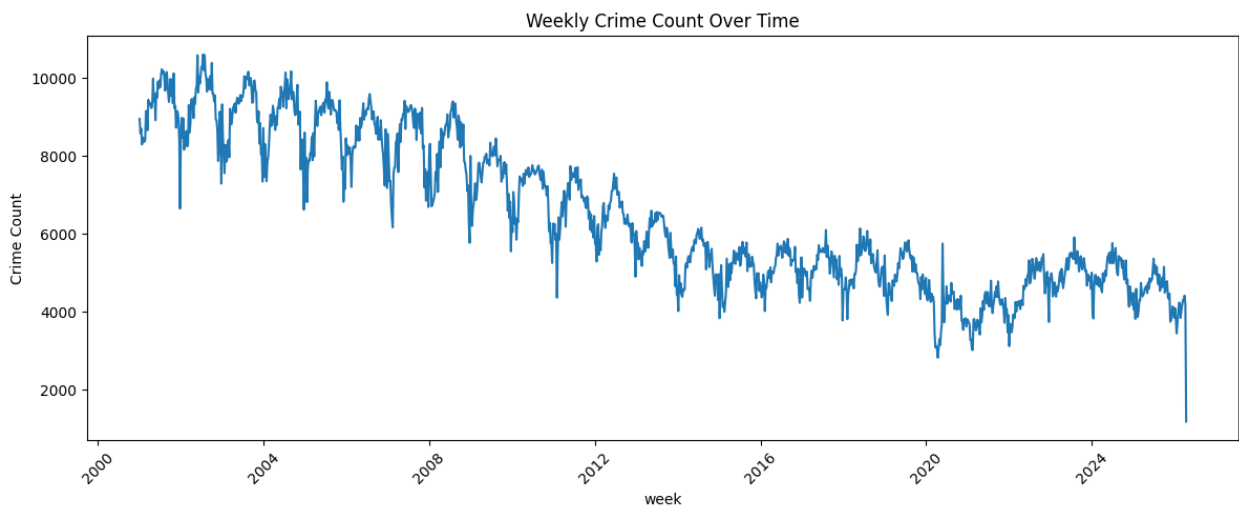
The boxplot of the crime counts shows the central focus and spread of the data. This boxplot shows that the median crime count is approximately 5800 crimes and the interquartile range is between 4900 and 8200, indicating that about 50 percent of the weeks are in this range. The minimum number of crimes is 2800, and the maximum is 10600 crimes, which shows a large range across

all weeks. Besides that, this box plot shows that data is right skewed, which presents that some weeks have unusually high crime counts. This boxplot also emphasized that there is no outlier in this dataset (Figure2).

Figure 2



Also, Figure 3 presents the time series plot of crime counts over time. This plot has a temporal structure that has a decreasing trend and includes seasonal peaks.



In the next step, we needed to prepare the dataset for modeling. Therefore, we normalized the data by subtracting the mean and dividing by the standard deviation. In the next stage, we use a sliding window approach to transform a single continuous sequence into multiple input and output pairs. So, each input includes 500 past observations as we set the length of the sequence to 500, and the related output is the next value, which follows the generated sequence. By using this method and

moving the window across the whole dataset, we could generate many training samples. These samples were then used to train the model and predict future values due to past observations.

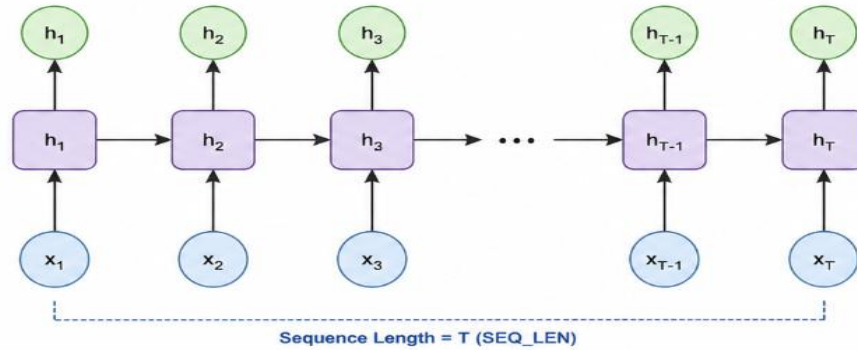
So, for modeling, we split our preprocessed time series dataset into a train set and test set by dedicating 80 percent of the dataset to the train set and 20 percent to the test set. Then, we reshape it into an essential format that can be used for RNN models such as LSTMs and GRUs. Then we developed various types of RNN models to evaluate their performance for prediction.

Model Development

Our data is time-series data with seasonal patterns. Therefore, we need to use appropriate models that can capture sequential dependencies for forecasting future values, and these models are recurrent neural network models. Our dataset has long sequential data, and we could just focus on LSTM and GRU models that have memories and are more common for long sequential datasets. However, we tried to develop different RNN models to perform evaluation on all of them and introduce the best one. Also, each of these models has their own strengths and limitations.

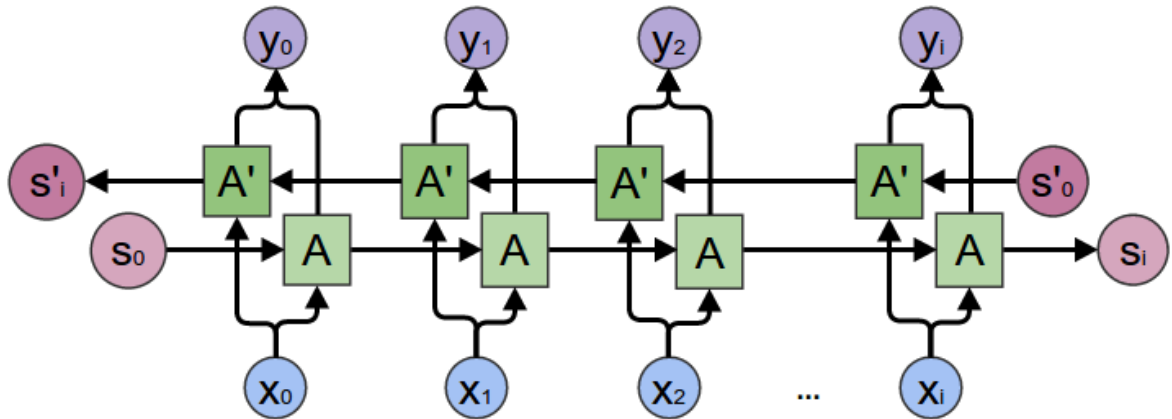
Vanilla RNN model

The first model that we used was a vanilla RNN model. This model is a good choice for fitting on time series data, and it is an efficient and fast processing model, but it is memory-free, and it cannot deal with vanishing gradients and has limitations in learning long-term patterns. So, if the number of time steps increases, it cannot have good performance in forecasting. The one that we built has 32 hidden units. The Vanilla Recurrent Neural Network (RNN) model developed in this study has 32 hidden units, and it has a dense output layer with one neuron. This model takes the input with length equal to 500 and uses the Tanh activation function. When all the sequences are processed, the final hidden state is fed to the dense layer and then creates a single output value that shows the predicted value for the next time step. The Adam optimizer was also utilized to train this model, and 10 percent of the training data was set apart for validation. Therefore, in the training process, the model learns the weights and tries to update them by using backpropagation.



Bidirectional Model

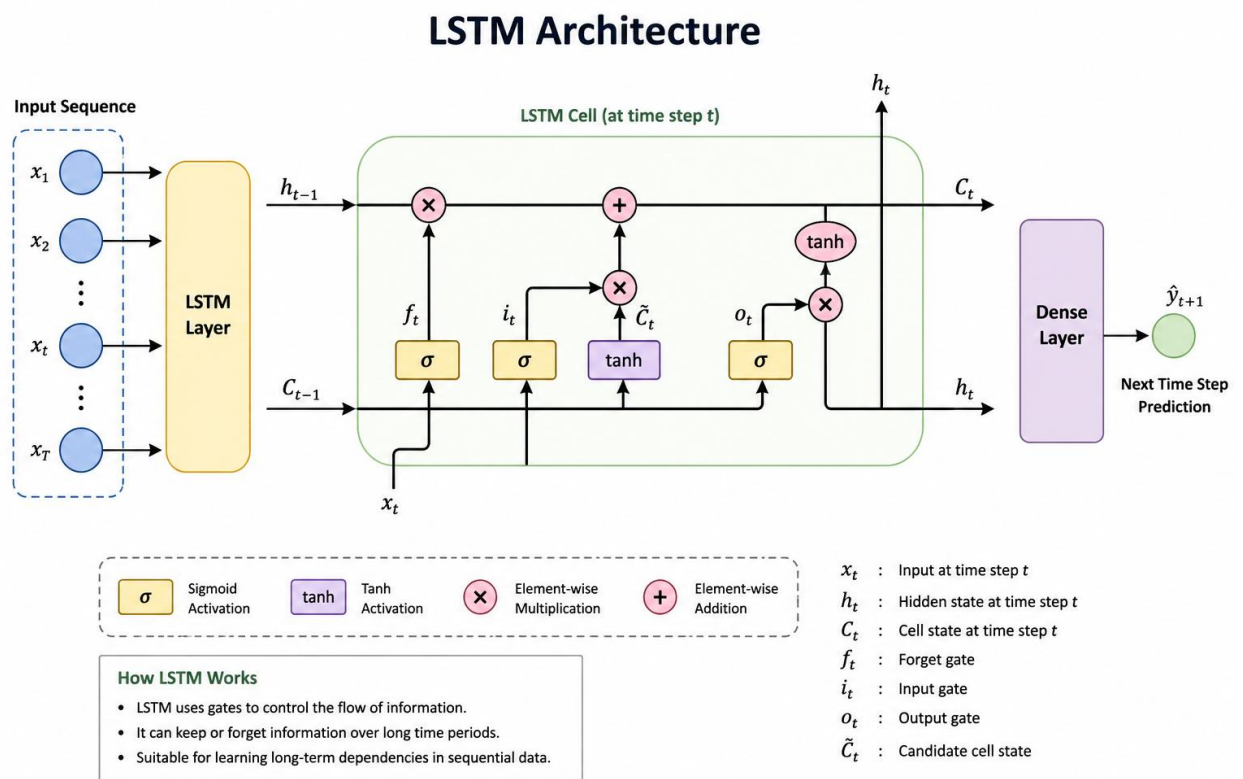
The bidirectional RNN model is the extended form of the vanilla RNN, and it tries to cover some limitations of the vanilla RNN model. This model processes the sequences from both sides, including forward and backward; therefore, it can deal with vanishing gradients very well. The hidden states from both directions are combined at each time step, with each of these hidden states having 32 neurons, and use the Tanh activation function. The combined output of the hidden layers is then passed to a dense layer and mapped to a single numeric prediction. In the training process the model learns the parameters of both forward and backward RNNs by considering mean square error as a loss function and using the Adam optimizer.



Long Short-Term Memory (LSTM) Model

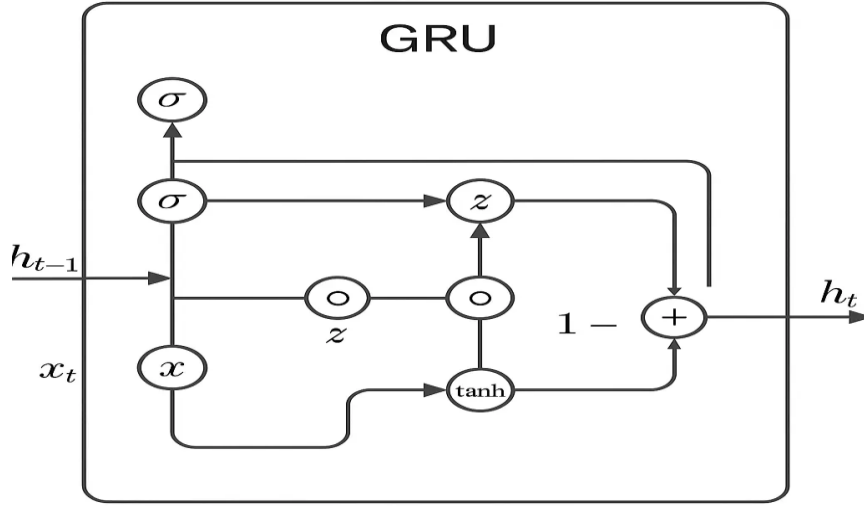
We also tried to develop more advanced models, such as Long Short-Term Memory, or LSTM, that can address the limitation of vanishing gradients. This type of model has a gating mechanism that helps to keep or forget information over long sequences. Therefore, LSTM is very appropriate for capturing long-term dependencies in time series data. So, due to our dataset, it can learn periodic peaks and decreasing trends very well. However, LSTM is a complex model, and its training needs more time compared to simpler forms of RNN models. An LSTM model contains

an input gate, an output gate, and a forget gate that help to decide which information should be kept and which should be discarded. So these gates help to just keep important information over long sequences. This model has two important components, including hidden state and cell state. The model, which was built in this study, used the Tanh activation function, the Adam optimizer, and mean square error as the loss function.



Gated Recurrent Unit (GRU) Model

Similar to LSTM, the Gated Recurrent Unit, or GRU, is also an appropriate model for dealing with vanishing gradients. This model also has memory that helps keep the previous information to generate outputs using the reset gate and update gate. The GRU model has a simpler structure and can be trained faster with similar performances compared to the LSTM model.



ARIMA Model

In addition to deep learning models, we also deploy one traditional statistical model that especially was used widely in the literature. This model is very effective for modeling linear relationships and trends. This model is not suitable for nonlinear relationships that especially have multiple seasonal behaviors, which deep learning models can handle effectively. So, we developed it just for comparison with other models and because it was used in different previous papers.

In this study, we basically use a classical AutoRegressive (AR) model, which is a simple form of ARIMA. It starts with first-order differencing and transforms the original series into differences between consecutive values. This helps remove the trends and make data more stable. Then we created input-output pairs by using a sliding window. The model was built by using the ordinary least squares method, and the model tries to learn a set of coefficients, for example, the linear combination of the previous values to predict the next value. Then for making the predictions meaningful, we integrated back the differencing by adding the predicted differences back to the starting values.

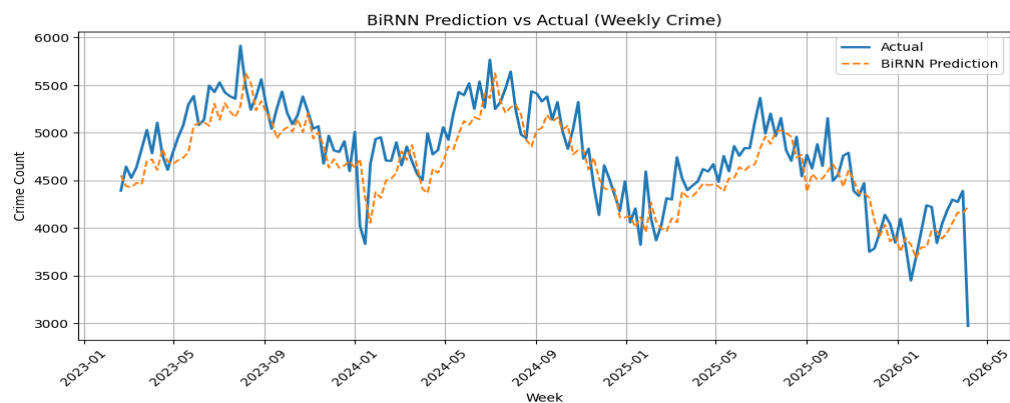
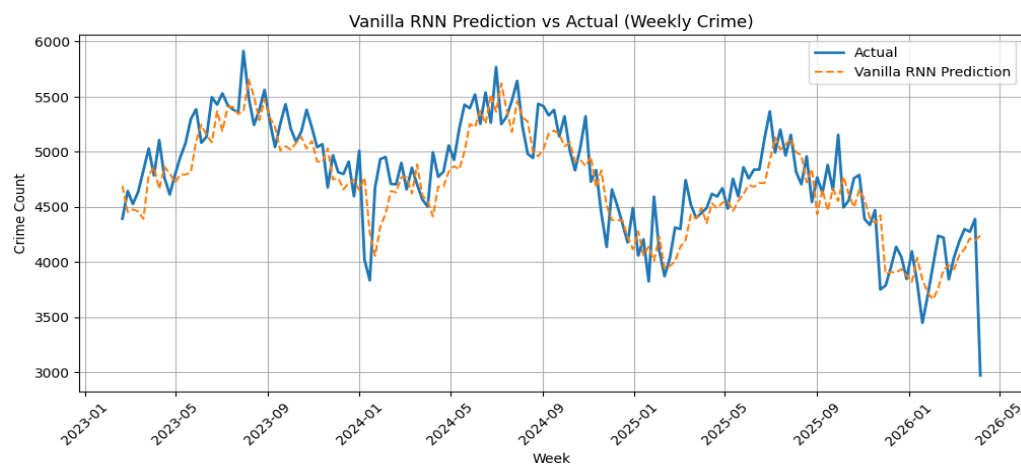
Evaluation Metrics and hyperparameters

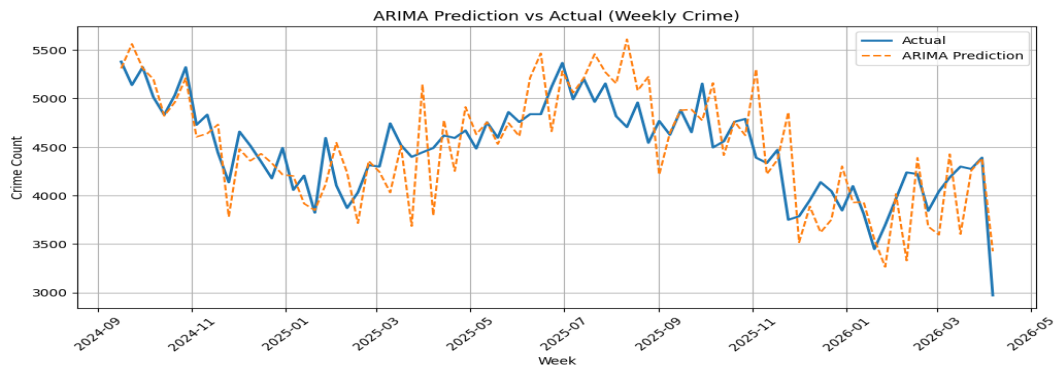
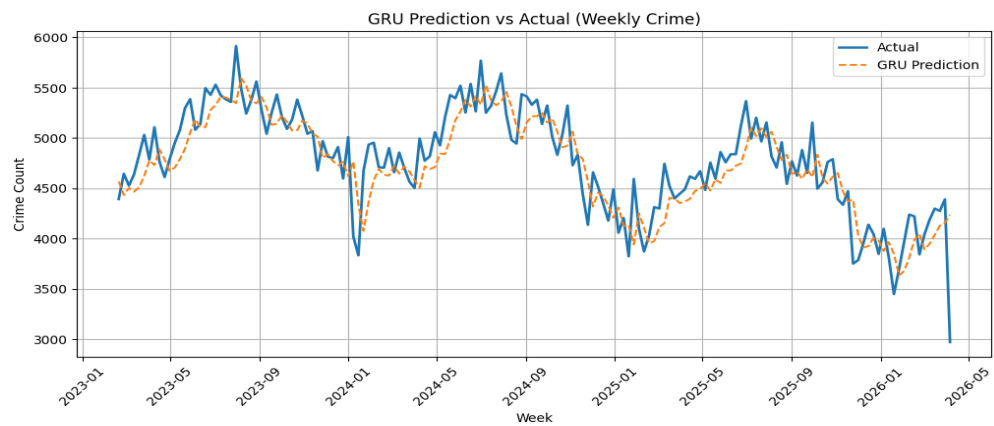
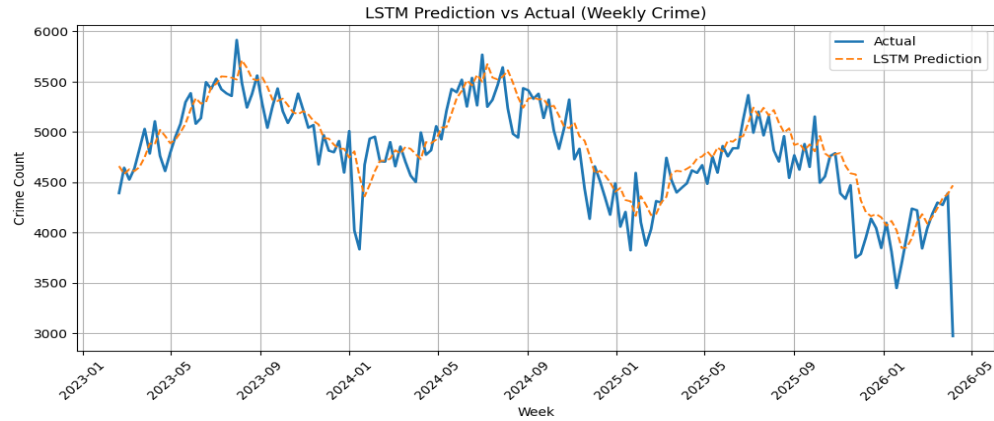
In this study, we used different evaluation metrics for assessing the performance of the models, including root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), which makes the error interpretation easier. Finally, R-squared, which measures how well the model explains the variance in the data.

A total of 32 neurons were used for creating hidden layers, and all RNN models were trained using 50 epochs with a learning rate equal to 0.001 and a batch size equal to 16.

Results and Discussions

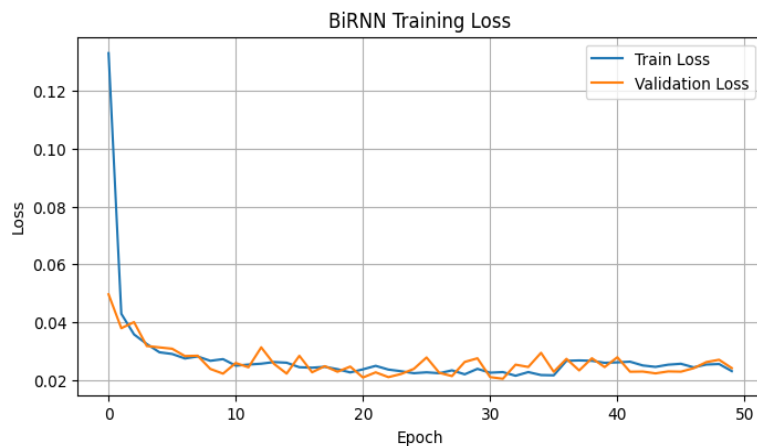
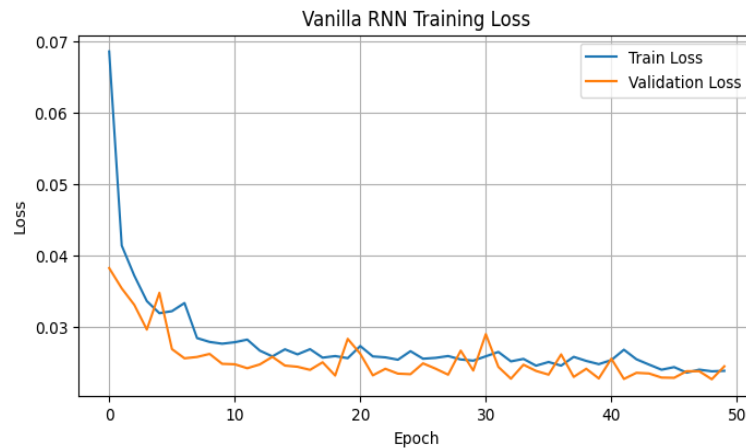
Finally, for visualizing the results, we create the plots of actual versus predicted values to see the performance of the model across all time steps. The comparison of model predictions against the actual weekly crime data shows their differences in performance. The RNN models could capture the general trends and seasonality of the time series data, and their plots show the upward and downward patterns similar to the actual data; however, there are significant differences in accuracy. The GRU model would have the best performance and the predictions aligned more closely with the actual values. The LSTM model also shows strong performance, and it could track the actual series but with less precision than GRU. In contrast, the ARIMA model performs significantly worse than RNN models. Their predictions are inconsistent and do not align with actual data.

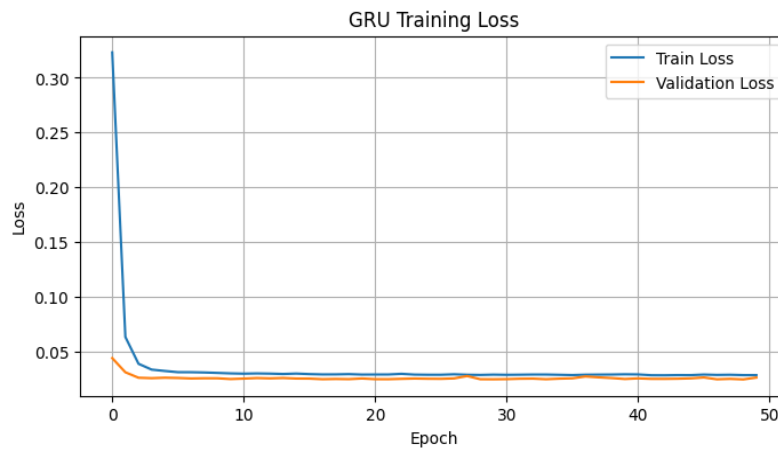
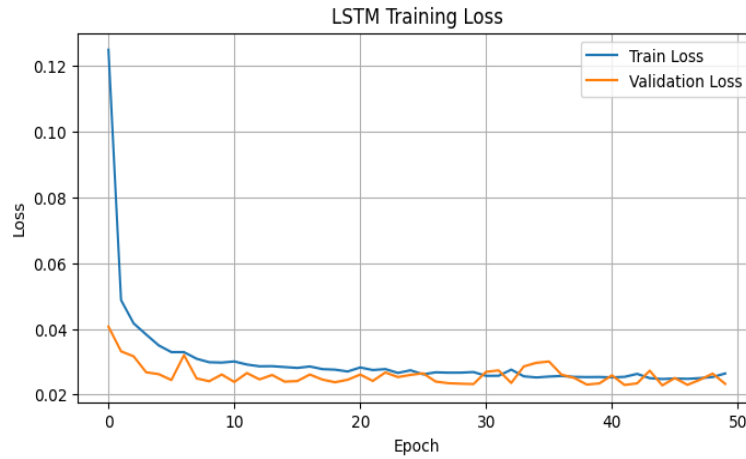




We also visualize the training and validation loss for all deep learning models, including RNN, BiRNN, LSTM, and GRU. These plots present stable and effective learning behavior. All models showed a rapid decrease in loss during the first epochs, and then they became relatively stable. These demonstrated that all models can learn the patterns in data very well. The GRU model shows the fastest convergence with a sudden decrease in training loss in the initial epochs, and it has minimum oscillation after that. After GRU, the LSTM model could also show stable convergence

that presents that it can be generalized very well and without overfitting. Similarly, vanilla RNN shows a good learning process and shows minor fluctuations in comparison to GRU and LSTM. BiRNN also shows a similar trend, but it begins with a high initial loss because of its more complex structure and converges to a higher loss compared to others. Since the training and validation loss are very close to each other for all models, it means that they could generalize well and do not show any signs of overfitting.





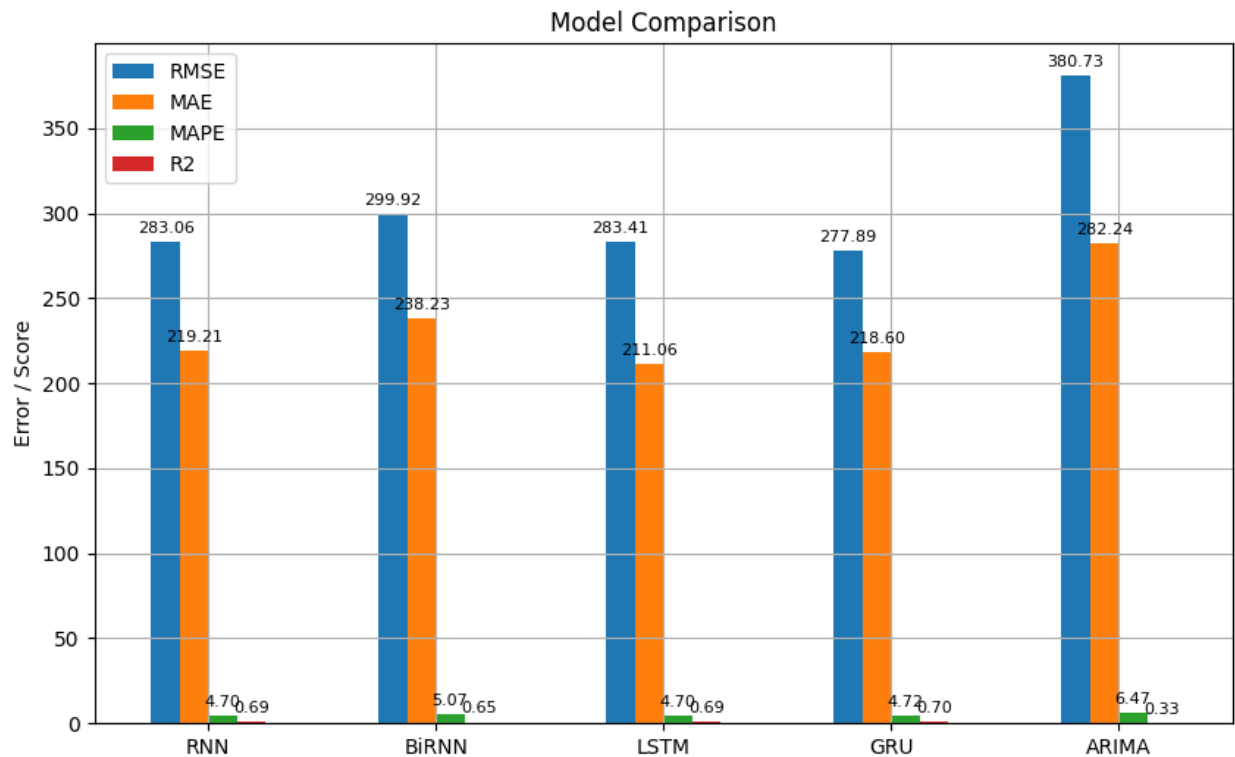
Performance comparison of the five models presented that deep learning models had better performance than traditional statistical models. Among all models, GRU could reach the best performance with the lowest Root Mean Square equal to 277.89 and the highest R-square equal to 0.70. This means that this model could provide the most accurate predictions and could explain the large part of the variance in the data. The LSTM model and Vanilla performed similarly, and they have RMSE around 283 and R-squared values close to 0.69. They also demonstrated strong predictive performance. LSTM could perform better than RNN by focusing on MAE, as it could achieve 211 versus 219. It presents that it has better average prediction accuracy. The BiRNN presents weaker performance in comparison to these models with a higher RMSE (299.92), MAE (238.23), and lower R-squared (0.65). So, using BiRNN could not present any additional benefit.

In contrast, ARIMA has the worst performance compared to all RNN models, with the highest RMSE (380.73), MAE (282.24), and lower R^2 (0.3298). This model shows poor predictive

accuracy and low ability to explain the variability in the data. The values for all the metrics of these models were shown in table 1

Model	RMSE	MAE	MAPE	R ²
RNN	283.0609	219.2085	4.7042	0.691
BiRNN	299.9187	238.2329	5.07	0.6531
LSTM	283.4071	211.0592	4.6977	0.6903
GRU	277.8887	218.598	4.7207	0.7022
ARIMA	380.7254	282.2438	6.4673	0.3298

We also illustrate the comparison of these models in terms of different metrics by visualizing them in a bar chart. GRU presents the best performance compared to all other models.



Advantages and Disadvantages of Time Series Models:

Model	Advantages	Disadvantages
Vanilla RNN (Recurrent Neural Network)	* Good for fitting time series data. * Fast and efficient processing.	* Does not hold memory. * Cannot deal with vanishing gradients. * Limited in learning long-term patterns. * Performance suffers as the number of time steps increases.
Bidirectional RNN (Bi-RNN)	* Extended form of Vanilla RNN. * Helps to process sequences from both forward and backward directions. * Can deal with vanishing gradients.	* More complex structure, resulting in a high initial loss during training. * Converges to a higher loss compared to other models. * Presented weaker performance in this study compared to RNN, LSTM, and GRU. * Did not present any additional benefit in this study.
LSTM (Long Short-Term Memory)	* Addresses issues with vanishing gradients. * Gating mechanism helps to keep or forget information over long sequences. * Appropriate for capturing long-term dependencies in time series data. * Can learn periodic peaks and decreasing trends very well.	* Is a complex model. * Training needs more time compared to simpler forms of RNN models. * Can be more expensive than GRU models

GRU (Gated Recurrent Unit)	* Addresses issues with vanishing gradients. * Has memory to store previous information for the outputs. * Simpler model and more time efficient in comparison to LSTM models. * Fastest convergence with minimum oscillation during training.	* None explicitly stated in the context.
ARIMA (Autoregressive Integrated Moving Average)	* Effective for modeling linear relationships and trends. * Traditional statistical model that was used widely in the literature.	* Not suitable for nonlinear relationships, such as multiple seasonal behaviors. * Had the worst performance compared to all RNN models (highest RMSE, lowest R^2). * Shows poor predictive accuracy and a low ability to explain the variability in the data.

Conclusions

This project tries to create a pipeline including data preprocessing and transforming it into weekly time series as well as training different models and using different evaluation metrics for the prediction of weekly crime count in Chicago. The major contribution of this project was developing and comparing multiple RNN models for performing weekly forecasts of crime counts using a 20-year dataset from Chicago. We designed and assessed the vanilla RNN, bidirectional RNN, LSTM, and GRU along with the traditional ARIMA model. Our findings presented that deep learning models, especially the GRU model, have better performance in comparison to other RNN models and the ARIMA model. GRU could perform well in comparison to others and also LSTM because it can keep memory from past data, and also it needs a lower number of parameters in comparison to LSTM and has a simpler structure. The results of this project can be beneficial

for different stakeholders, including law enforcement agencies, to help them optimize allocation of resources and improve the response time and especially manage high-risk periods. Policy makers and city planners can also take advantage of this pipeline by applying these insights for targeted interventions and allocating public safety resources. Data scientists and researchers can use these results for time-series forecasting and model comparisons. The residents of Chicago can benefit from these results by having improved safety measures.

For future research, we can incorporate additional features such as geographic locations, socioeconomic factors, or demographic data to improve the model accuracy. We can apply transformers to enhance the performance of the model, and apply multi-step forecasting and prediction for multiple weeks. The findings of this study also have significant implications across different domains, such as public safety to plan for crime prevention strategies and urban planning for resource distribution. In fact the informed decisions made by these models can improve urban efficiency.

Furthermore, this project has also economic value, as these models can help to predict future weekly crimes and therefore lead to significant reduction of costs associated with crime and inefficient resource allocation. These reductions in expenses will happen by improving the prediction accuracy, more effectively using the personnel, and preventing crimes that cause an increase in property damage and healthcare costs. So, by making the public safety systems prepared and resilient, we can have significant long-term economic benefits.

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